Supplement S5: Cached-Qwen search follow-up

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1 Evidence chronology and scope

The 10 November editorial report summarizes searches completed on 9 November; it is not a new run on 10 November. The 11 November OPT-only check is a distinct diagnostic (finite CPU Slurm job 27702). Both reuse cached forecasts and exposed development respondents. They search mixture parameters rather than generating or optimizing new prompt text. Neither accesses the proposed independent external panel's TEST outcomes. These results do not update the original search-family or R2 Holm decisions.

2 TRAIN320 robustness

The frozen TRAIN seeds are 20261110, 20261111 and 20261112; search seeds are 11, 29 and 47. Each statistical fit uses TRAIN-only three-fold selection among prior, ridge and CatBoost configurations; all three select CatBoost depth 4, 100 iterations, learning rate .05. Existing OPT80 and SELECT120 select candidates before the common exposed DEVTEST400 evaluation. CMA-MAE and random each have a cap of 70 evaluations; all 18 searches complete (1,260 evaluations). The common simple comparator is selected on SELECT, separately within each fit. TRAIN320 counts only fitting of the statistical component, not all historical supervision. Training subsets overlap by 8/11/11 people and share the evaluation cohort.

TRAIN seed	Statistical RPS	Simple RPS	Median CMA minus simple
20261110	.12063438	.11943963	-.00069552
20261111	.11987651	.11883056	-.00056003
20261112	.11987067	.11860234	-.00033367

The median of fit-level median differences is $-.00056003$ against simple and $-.00017157$ against random. CMA-MAE wins eight of nine simple comparisons (one tie) and seven of nine random comparisons (one tie, one loss). The fixed practical rule requires median difference to simple $\leq -.001$, no seed loss to simple, negative median difference to random and at least two random wins per three seeds. Zero of three fits passes; broad continuation required all three. This is a screening decision, not an equivalence test or evidence of universal optimizer failure. The original n320 fit's median simple-comparator gain of .000795 also falls below .001. Earlier positive n80 evidence is retained as historical evidence: these differing designs do not isolate a budget effect.

3 All matched TRAIN320 cells

TRAIN seed	Engine	Search seed	RPS	Difference to simple
20261110	random	11	0.11882514	-0.00061449
20261110	random	29	0.11877111	-0.00066852
20261110	random	47	0.11943963	+0.00000000
20261110	cmamae	11	0.11864103	-0.00079860
20261110	cmamae	29	0.11874411	-0.00069552
20261110	cmamae	47	0.11943963	+0.00000000
20261111	random	11	0.11874946	-0.00008109
20261111	random	29	0.11831136	-0.00051919

20261111	random	47	0.11883056	+0.00000000
20261111	cmamae	11	0.11821682	-0.00061373
20261111	cmamae	29	0.11827052	-0.00056003
20261111	cmamae	47	0.11865898	-0.00017157
20261112	random	11	0.11803562	-0.00056671
20261112	random	29	0.11790844	-0.00069389
20261112	random	47	0.11860234	+0.00000000
20261112	cmamae	11	0.11783148	-0.00077085
20261112	cmamae	29	0.11841423	-0.00018811
20261112	cmamae	47	0.11826867	-0.00033367

4 OPT80 optimizer contrast

The fixed existing n320 predictor is retained. A seeded permutation (2026111101) partitions OPT80 into three groups of 27, 27 and 26. Rotating optimization, calibration and validation roles gives three disjoint validation groups but overlapping role assignments across rotations. Each engine uses the same four gate coefficients bounded to $[-6, 6]$, six common simple anchors included in the 70 evaluations, and a shortlist containing simple candidates plus the five best adaptive gates on optimization loss. Calibration selects the winner before validation scoring. Three seeds (11,29,47) and three engines produce 27 cells. Alternative energy-form RPS agrees with validation losses within the configured numerical tolerance; this is a scorer check, not independent scientific replication.

Engine	Mean RPS	Mean difference to simple
CMA-MAE	.118534815	-.000719275
Random	.118718431	-.000535659
Coordinate	.118590691	-.000663399

Seed	CMA minus random	CMA minus coordinate
11	+.000037775	-.000326418
29	+.000288444	+.000132817
47	-.000877067	+.000025972

The mean seed-level differences are $-.000183616$ and $-.000055876$, respectively, with only one win out of three against each rival. These summaries average losses over the 80 validation people per seed; folds are not independent cohorts. We report no confirmatory confidence interval or p-value. The OPT-only diagnostic is not another application of the TRAIN320 continuation rule or a fresh holdout. Historical cache-generation and statistical-fitting costs are excluded from the optimizer-only work; neither diagnostic establishes total-cost efficiency. The underlying data and individual predictions remain restricted. Aggregate summaries and all TRAIN320 search rows accompany this revision.

5 All OPT80 development cells

Fold	Seed	Engine	RPS	Difference to simple
0	11	cmamae	0.122298091	-0.000218961
0	11	random	0.121637941	-0.000879111
0	11	coordinate	0.122054552	-0.000462500
0	29	cmamae	0.122053390	-0.000463662
0	29	random	0.121124489	-0.001392563
0	29	coordinate	0.121417093	-0.001099959
0	47	cmamae	0.121380474	-0.001136578
0	47	random	0.122517052	+0.000000000
0	47	coordinate	0.121280464	-0.001236587
1	11	cmamae	0.112570759	-0.001076548

1	11	random	0.113072418	-0.000574889
1	11	coordinate	0.113657065	+0.000009758
1	29	cmamae	0.112734756	-0.000912551
1	29	random	0.112762851	-0.000884456
1	29	coordinate	0.112983080	-0.000664227
1	47	cmamae	0.113448397	-0.000198910
1	47	random	0.113647307	+0.000000000
1	47	coordinate	0.113299071	-0.000348236
2	11	cmamae	0.120773066	-0.000914993
2	11	random	0.120821421	-0.000866638
2	11	coordinate	0.120902248	-0.000785811
2	29	cmamae	0.120890769	-0.000797290
2	29	random	0.120938703	-0.000749356
2	29	coordinate	0.120884996	-0.000803063
2	47	cmamae	0.120922213	-0.000765846
2	47	random	0.122234026	+0.000545968
2	47	coordinate	0.121101223	-0.000586836